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ORIGINAL ARTICLE

Isolation and antimicrobial resistance of motile *Salmonella enterica* from the poultry hatchery environment

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Abstract
Salmonella is a globally distributed major food-borne pathogen and poultry is one of the predominant sources of salmonellosis in humans. To investigate the presence of motile *Salmonella* in the poultry hatchery environment, we collected 97 fluff samples from four selected broiler breeder chicken hatcheries from Chattogram, Bangladesh during July-December 2015. To isolate motile *Salmonella enterica*, we used conventional bacteriological techniques followed by serological verification using anti-*Salmonella* Poly A-E serum and species confirmation by conventional PCR assay. Antimicrobial susceptibility testing by Kirby-Bauer disc diffusion method for 10 commonly used antibiotics was performed on all isolates. Isolates displaying phenotypic resistance to ampicillin were tested by PCR for bla_{TEM} gene, whereas those resistant to tetracycline were tested for the presence of *tet*A, *tet*B and *tet*C genes. A total of 24 samples (24.7%; 95% CI: 16.5–34.5, N = 97) from 3 hatcheries were positive for motile *Salmonella*. Of them, 21 (87.5%) and 12 (50.0%) were resistant to ampicillin and tetracycline, respectively, 9 (37.5%) to nalidixic acid and sulphamethoxazole/trimethoprim. No resistance was detected to ceftriaxone, cefoxitin, gentamicin, neomycin, ciprofloxacin and colistin. Ten (42%) of 24 isolates from 2 hatcheries were multi-drug resistant (i.e. resistant to ≥ 3 antimicrobial classes). Six of 21 ampicillin resistant isolates contained bla_{TEM} gene and 10 of 12 tetracycline resistant isolates contained *tet*A gene. This study highlights the circulation of multi-drug resistant motile *Salmonella* in the hatchery environment for the first time in Bangladesh. Further epidemiological and molecular studies are therefore needed to identify the serotypes and source of the bacteria in hatcheries.

Keywords Bangladesh · bla_{TEM} · Food-borne pathogen · Multi-drug resistant · *Salmonella* · *tet*A · Zoonotic

Introduction

Non-typhoidal *Salmonella* (NTS) is a globally distributed major food-borne pathogen and poultry is a main reservoir of many non-host specific serovars of *Salmonella.* Poultry hatcheries can be an important source of *Salmonella* infection to poultry flocks (Crabb et al. 2018). Contaminated poultry products are frequently associated with human infection (Foley et al. 2011), which can be expensive for the poultry industry, governments, and affected individuals in terms of recall of eggs and poultry products, cost of treatments, hospitalization, income loss etc. The zoonotic nontyphoidal *Salmonella enterica* from developing countries like Bangladesh can easily be spread to other countries across the world through international travel, human migration, food, animal feed and poultry trade (Aarestrup et al. 2007; Collard et al. 2007). Zoonotic *Salmonella* should be eliminated from the source of origin to reduce the risk of wide scale transmission. To reduce the prevalence of *Salmonella* in poultry and consequently the burden of human salmonellosis, the European Union introduced and subsequently implemented a monitoring and control program in all poultry production systems across the member countries (EFSA 2011). However, no monitoring system currently exists in Bangladesh due to resource constraints.

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Poultry acts as reservoir of motile *Salmonella enterica*. *Salmonella enterica* is divided into six subspecies and each sub-species contains multiple serotypes. More than 2600 serotypes belonging to *Salmonella enterica* sub-species have been identified (Cheng et al. 2019). However, only two non-motile serovars (Gallinarum and Pullorum) cause disease in chickens while most other serovars are motile and can infect humans (Gast et al. 2020). Many of these serotypes are found in poultry without causing any clinical illness (Kabir 2010). The most prevalent serotypes under sub-species *enterica*, responsible for causing infections in humans and animals are *Salmonella* Enteritidis, *Salmonella* Typhimurium and *Salmonella* Heidelberg (Gast et al. 2020).

Salmonella can be introduced into poultry flocks from infected hatcheries. The same *Salmonella* serotypes responsible for infection of chicks are often found in their parent flocks (Crabb et al. 2018). Thus, *Salmonella* infection in newly hatched chicks is frequently a cause for infections in mature commercial layer and broiler flocks (Namata et al. 2009; Marín et al. 2011). Any salmonellae carried in or on eggs can spread extensively in hatcheries. As chicks pip through eggshells, salmonellae are released into the air and circulated around hatching cabinets on contaminated fluff and other hatchery debris (Volkova et al. 2011). Therefore, controlling *Salmonella* in the hatchery is the first step to prevent dissemination into the subsequent stages of poultry production.

Antimicrobial resistant of NTS have been emerged worldwide. Several studies have reported ampicillin, amoxicillin and trimethoprim-sulfamethoxazole resistance, particularly among the globally prevalent serotypes of *Salmonella* Enteritidis and *Salmonella* Typhimurium (Meakins et al. 2008; Crump et al. 2011; Odoch et al. 2018). Of particular concern is the emergence of extended spectrum beta-lactamase genes in plasmids of NTS (Denagamage et al. 2019), as well as reports of carbapenemase-containing NTS isolates (Miriagou et al. 2003; Nordmann et al. 2008), both of which confer resistance to highly important antimicrobials. Fluoroquinolones and third generation cephalosporins are frequently used for the treatment of NTS in humans that are resistant to conventional antibiotics, although reports of increasing resistance to these critically important agents are emerging (Parry and Threlfall 2008).

Motile *Salmonella* has been identified in layer, breeder and commercial broiler poultry farms in Bangladesh (Barua et al. 2012, 2013), but the primary source of motile *Salmonella* in layer and breeder farms has not yet been explored. The present study – i.) determined the prevalence and antimicrobial resistance of motile *Salmonella enterica* in the poultry hatchery environment, and ii.) investigated the presence of some selected antimicrobial resistance genes in the multidrug resistant *Salmonella* isolates.

Materials and Methods

Study design, description of hatcheries and population

We conducted the study on four purposively selected broiler breeder chicken hatcheries (ID: A, B, C and D) in Chattogram, Bangladesh between July and December 2015. All the hatcheries reared Cobb-500 broiler parent stock. The average production of these hatcheries ranged from 45,000 to 85,000 day-old-chick per week. Day-old-chicks were distributed from these hatcheries to multiple locations within Bangladesh.

Collection of samples

Hatcher fluff samples were collected for isolation of *Salmonella*. Fluff is a potential source of *Salmonella* transmission and it has been used for monitoring *Salmonella* in many countries although some other samples from poultry hatcheries, e.g., broken eggshells, hatcher basket liners can also be used for *Salmonella* isolation (Carrique-Mas and Davies 2008). Contaminated fluff may infect early hatching embryos in the incubator, but also occurs during pipping, hatching in the hatchers, or during transportation of chicks to farms (Cox et al. 1991).

We collected 97 pooled fluff samples from the four hatcheries. Each pool consisted of 8–10 cross-sectional pinches for obtaining a total volume of approximately 200 gm. Hatchery A was physically visited eight times, whereas hatchery B and C were visited three times to collect the samples. There was at least 15 days interval between each visit to hatchery. During each visit, we collected samples from different hatch of parent stocks. Samples were sent via a personal messenger from hatchery D once. As hatchery A was near to the testing laboratory, we collected a higher number of samples from that hatchery (n = 72). The other three hatcheries were outside of the city area and due cost associated with the sampling those hatcheries, a smaller number of samples were collected (n = 25; B = 15, C = 6 and D = 4).

Each pooled fluff sample was placed in sterile plastic zipper bags with a unique identity number. Samples were then transported to the microbiology laboratory, Chattogram Veterinary and Animal Sciences University (CVASU), Bangladesh within two hours. On arrival at the laboratory, the samples were stored at 5 °C overnight, if they were not able to be processed on the same day as sample collection.

Isolation and identification of Salmonella

Salmonella was isolated from pooled fluff samples by conventional bacteriological procedure. Briefly, 25 gm of each pooled fluff samples were transferred to a conical flask or blue-capped bottle containing 200 ml buffered peptone water (BPW; Oxoid Ltd., UK) and mixed using a vortex machine to obtain a homogenous suspension. Subsequently the flasks were incubated aerobically at 37 °C for 18 h. Then 100 µl of the BPW culture, divided into three separate and equally spaced drops, were inoculated on to modified semi-solid Rappaport Vassiliadis (MSRV) agar (Oxoid Ltd., UK) supplemented with novobiocin (Oxoid Ltd., UK) and incubated aerobically at 41.5 °C for 24–36 h. If there was any swarming growth observed on the MRSV plates, an inoculating loop was dipped into the swarm zone and streaked on to the surface of brilliant green agar (Oxoid Ltd., UK) and incubated overnight at 37 °C. Suspected *Salmonella* colonies were transferred onto blood agar and incubated at 37 °C for 24 h. The cultures were identified biochemically by using triple-sugar iron agar. Following biochemical identification, well isolated 4–5 colonies from blood agar plate were picked up by inoculating loop and inoculated into 10 ml of nutrient broth (Oxoid Ltd., UK). The inoculated nutrient broths were incubated overnight at 37 °C and then 700 µl of culture was transferred to cryovial containing 300 µl of 50% glycerol and stored at -80 °C until further testing.

Isolates showing typical biochemical reactions were then confirmed serologically by agglutination test using anti-*Salmonella* polyvalent A-E serum (Statens Serum Institut, Copenhagen, Denmark). Each suspected *Salmonella* culture from blood agar plates was mixed with a drop of polyvalent antisera and incubated for up to 2 min at room temperature. Phosphate buffered saline (PBS) was used as a control to check for auto agglutination of the individual isolate.

Genomic DNA of serologically confirmed *Salmonella* isolates was extracted by the boiling method (Gouws et al. 1998) and a *Salmonella* species specific PCR assay was conducted for the final identification of isolates (Gouws et al. 1998) using the set of primers described in Table 1. PCR products on the 1% agarose gel were visualized through ethidium bromide staining (Fig. 1). Isolates which were showing typical biochemical reactions, positive in both agglutination tests and in PCR, were considered as positive for motile *Salmonella enterica.*

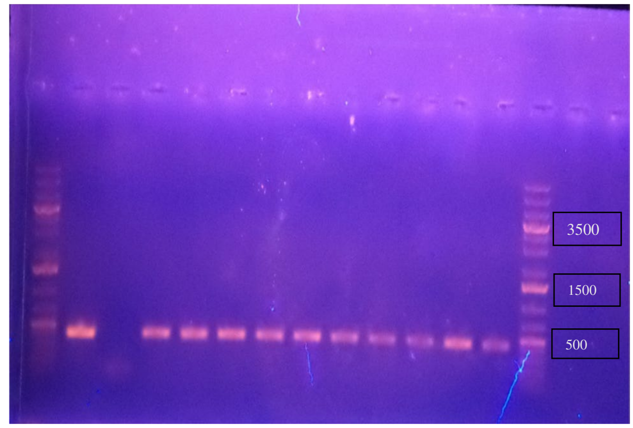

Fig. 1 Amplicons detected at the level of 429 bp size region against 1 kb + DNA ladder

Antimicrobial susceptibility testing

All *Salmonella* isolates were tested for their in vitro sensitivity to 10 commonly used antimicrobials using disk diffusion method following the guidelines described by Clinical and Laboratory Standards Institute (CLSI) (CLSI 2018). The following antimicrobials were tested: ampicillin (10 µg), cefoxitin (30 µg), ceftriaxone (30 µg), ciprofloxacin (5 µg), colistin sulphate (10 µg), gentamicin (10 µg), nalidixic acid (30 µg), neomycin (30 µg), sulfamethoxazole-trimethoprim (25 µg) and tetracycline (30 µg). The results of susceptibility testing were interpreted according to the criteria described by CLSI (2018). The *Escherichia coli* ATCC 25,922 was used as a quality control. An isolate was defined as 'multi-drug-resistant (MDR)' if it displayed resistance to ≥ 3 different classes of antimicrobial (Tenover, 2006).

Table 1 Primer sequences for detection of *Salmonella* and antimicrobial resistance genes

Gene	Primer name	Primer sequence (5′–3′)	Annealing (°C)	Fragment size (bp)	References
	ST11	AGCCAACCATTGCTAAATTGG CGCA	60	429	(Olsen et al. 1999)
	ST15	GGTAGAAATTCCCAGCGGGTA CTG	60	429	(Olsen et al. 1999)
tetA	TetA-L	GGCGGTCTTCTTCATCATGC	64	502	(Lanz et al. 2003)
	TetA-R	CGGCAGGCAGAGCAAGTAGA			
tetB	TetB-L	CATTAATAGGCGCATCGCTG	64	930	(Lanz et al. 2003)
	TetB-R	TGAAGGTCATCGATAGCAGG			
tetC	TetC-L	GCTGTAGGCATAGGCTTGGT	64	888	(Lanz et al. 2003)
	TetC-R	GCCGGAAGCGAGAAGAATCA			
bla_{TEM}	757	GCGGAACCCCTATTTG	50	964	(Hasman et al., 2005)
	821	TCTAAAGTATATATGAGTAAACTTGGTCTGAC			

Table 2 *Salmonella* isolation from the hatchery fluff samples collected from four different hatcheries in Chattogram, Bangladesh during July – December 2015

Hatchery ID	No. samples tested	Number Positive	Prevalence (%)	95% CI*
A	72	20	27.8	17.9–39.6
B	15	3	20	4.3–48.1
C	6	0	0	0.0–45.9
D	4	1	25	0.6–80.6
Total	**97**	**24**	**24.7**	16.5–34.5

*CI: confidence interval.

Detection of antimicrobial resistance genes

Salmonella isolates displaying resistance to tetracycline were tested for *tet*A, *tet*B and *tet*C genes, and those displaying resistance to ampicillin for bla_{TEM} gene by PCR described by Lanz et al. (2003) and Hasman et al. (2005), respectively (Table 1). Genomic DNA extraction was performed using boiling method as described by Gouws et al. (1998). PCR products on the 1% agarose gel were visualized through ethidium bromide staining.

Statistical analysis

The data were entered into MS-Excel and exported to STATA-13 (StataCorp, 4905, Lakeway Drive, College station, Texas 77,845, USA) for analysis. We considered a hatchery as positive if at least one fluff sample was positive. The prevalence of motile *Salmonella enterica* for each hatchery was calculated as the number of positive samples divided by number of samples tested and expressed as a percentage with 95% confidence interval (CI). We also calculated for each antimicrobial the percentage of isolates being susceptible, intermediate, and resistant over all *Salmonella* positive isolates.

Ethical statement

As we collected only fluff samples from chicken hatcheries, verbal consent from the authority of each hatchery was taken before collecting the samples from the hatcheries.

Results

Prevalence of Salmonella

We isolated *Salmonella enterica* from 24.7% (24/97) of samples (95% CI: 16.5–34.5). The isolation rate varied among the four hatcheries. The highest percentage of samples positive with *Salmonella* was detected in hatchery A (27.8%) (Table 2).

Antimicrobial susceptibility profiles

All *Salmonella* isolates showed resistance to at least one antibiotic, except one isolate which was sensitive to all the tested antibiotics. Resistance to ampicillin was the most

Table 3 Antibiogram of 24 *Salmonella* isolates cultured from fluff samples of poultry hatcheries, Chattogram, Bangladesh July – December 2015

Antimicrobial agents	Susceptibility pattern		
	Sensitive	Intermediate	Resistant
	n(%)	n(%)	n(%)
Penicillin			
Ampicillin	3 (12.5)	0	21 (87.5)
Cephalosporins			
Ceftriaxone	23 (95.8)	1 (4.2)	0
Cefoxitin	24 (100)	0	0
Aminoglycosides			
Gentamicin	19 (79.2)	5 (20.8)	0
Neomycin	19 (79.2)	5 (20.8)	0
Tetracyclines			
Tetracycline	7 (29.2)	5 (20.8)	12 (50.0)
Quinolones and fluoroquinolones			
Ciprofloxacin	24 (100)	0	0
Nalidixic acid	4 (16.7)	11 (45.8)	9 (37.5)
Potentiated sulfonamides			
Sulphamethoxazole/Trimethoprim	15 (62.5)	0	9 (37.5)
Polymyxins			
Colistin	24 (100)	0	0

Table 4 Antimicrobial resistance patterns of *Salmonella* isolates (N = 23) cultured from fluff samples of different poultry hatcheries in Chattogram, Bangladesh, July – December 2015

Resistance phenotype*	No. of isolates showing resistance	Source (n)
AMP	5	Hatchery A (5)
AMP-TE	6	Hatchery A (6)
AMP-NA-TE	2	Hatchery A (2)
AMP-SXT-TE	1	Hatchery A (1)
AMP-NA-SXT-TE	7	Hatchery A (4), Hatchery B (3)
SXT	1	Hatchery A (1)
TE	1	Hatchery D (1)

*AMP = ampicillin, NA = nalidixic acid, SXT = sulphamethoxazole/trimethoprim, TE = tetracycline.

common (21/24; 87.5%) phenotype detected in the study, followed by tetracycline resistance (12/24; 50.0%). No resistance was detected to ceftriaxone, cefoxitin, gentamicin, neomycin, ciprofloxacin and colistin (Table 3).

Table 4 shows the phenotypic patterns of resistance among the *Salmonella* isolates. Seven isolates showed resistance to one antibiotic of which 5 (21.7%) isolates were resistant to ampicillin, 1 (4.3%) to sulphamethoxazole/trimethoprim and 1 (4.3%) to tetracycline. Six out of seven isolates originated from Hatchery A and the remaining one was from Hatchery D. Ten of 24 (42%) isolates displayed MDR, with seven (30.4%) resistant to four antimicrobial classes. The predominant resistance pattern was AMP-NA-SXT-TE.

Antimicrobial resistance genes

Ten of the 12 (83.3%) tetracycline resistant isolates contained the *tet*A gene. *tet*B and *tet*C genes were not detected in any of the tetracycline resistant isolates tested. Six out of 21 ampicillin resistant isolates (28.6%) gave positive amplicons for the bla_{TEM} gene.

Discussion

Presence of any non-host specific motile *Salmonella* in poultry is zoonotic and a public health concern in relation to food safety (Foley et al. 2011). Though information on prevalence and molecular characteristics of motile *Salmonella* from layer, broiler, and breeder farms are available (Barua et al. 2012, 2013), to the author's knowledge, this is the first report describing the presence of motile *Salmonella* in poultry hatcheries in Bangladesh. The study estimated a very high prevalence of motile *Salmonella enterica* in hatcheries (75%, N = 3 hatcheries). Similar prevalence has been detected in the United Kingdom (UK) in duck hatcheries (80%, N = 5) with several *Salmonella* serotypes including *S.* Typhimurium, *S.* Indiana, *S.* Senftenberg, *S.* Kottbus, and *S.* Bovismorbificans (Martelli et al. 2016).

The present study suggests persistent establishment of motile *Salmonella* in the hatchery environment. Overall, 24.7% (N = 97) of fluff samples were found positive with *Salmonella*. This is higher than a previous study (8.8%; N = 45 environmental samples) conducted by the United States Centre for Disease Control and Prevention (US-CDC) in a poultry hatchery following a human *Salmonella* outbreaks by *Salmonella* Enteritidis and *Salmonella* Litchfield in several states of the United States of America (USA) (Robertson et al. 2019). Moreover, *Salmonella* serovar prevalence was lower (8.7%) in fluff samples from broiler-breeder in Ontario, Canada (Sivaramalingam et al. 2013) than in our study. Samples collected from broiler production and distribution value chain (including hatcheries) of China showed that 10.6% (N = 11,592) of samples had *Salmonella* Enteritidis (Li et al. 2018). However, motile *Salmonella* serotypes were identified in 2.7–35.1% of poultry farms in South-Western states of Nigeria (Mshelbwala et al. 2017) and similar to our study 26.4% samples containing *S.* Typhimurium and *S.* Enteritidis in poultry farms of Pakistan (Wajid et al. 2019). All these variations in prevalence of motile *Salmonella* indicates that our management conditions align with other developing countries like Pakistan and Nigeria. If we want to reduce motile *Salmonella* level in hatcheries and subsequently in human food chain, we need to update our surveillance and control strategy.

Presence of *Salmonella* in sampled hatcheries indicates poor biosecurity and management. The major sources of *Salmonella* in hatcheries are incoming eggs (contaminated with feces, feather, and litter), people, rodents, insect vectors, and equipment (Bailey et al. 2001). Hatcher is the main source of contamination for the entire hatchery. Debris, dead chicks, and chick fluff are ideal replication sites for bacteria and fungi in the warm and humid hatchery environment (Animal and Plant Health Inspection Services-APHIS 2014) and this contamination is easily spread by air currents, equipment, and personnel (APHIS 2014). Attention to cleaning, disinfection, drying, and handling procedures is therefore of utmost importance to eliminate *Salmonella* infection.

Day-old chicks from the *Salmonella* contaminated hatcheries could contribute substantially to the introduction of *Salmonella* into broiler farms and consequently to retail outlets at the local live bird markets. Chickens brought from different farms share the same spaces in live bird markets and this may lead to the risk of *Salmonella* spreading from infected or carrier birds to healthy chickens and thereby cross contamination of carcasses during slaughtering and dressing. Previous human salmonellosis outbreaks were linked to live poultry contact, and infection can be traced back to hatcheries, highlighting the importance of reducing *Salmonella* at the hatchery level (APHIS 2014).

In this study, *Salmonella* isolates displayed resistance to a range of antimicrobials including quinolones. In absence of any monitoring system, the antimicrobial resistance temporal trend in this bacterium cannot be interpreted. Most of the isolates displayed resistance to ampicillin (87.5%) which corresponds to the report from another Bangladeshi study (82.9%) (Alam et al. 2020). In Bangladesh, most of the hatcheries inject single dose of ampicillin, amoxicillin, or gentamicin to day-old-chicks. This is done to reduce the chance of illness of the chicks prior to their distribution to farmers. Not only in hatcheries, but amoxicillin is also frequently used in layer chicken farms in Bangladesh (Imam et al. 2020). Therefore, the high level of resistance to ampicillin may reflect the injudicious use of these antimicrobials in the hatcheries. But we found sensitivity of motile *S. enterica* to gentamicin. Though this drug is commonly used in hatcheries, but rarely used in poultry farms. So, it is still effective against the studied bacterium.

Zoonotic *S. enterica* were resistant to nalidixic acid (37.5%) in this study. Study from Andoh et al. (2016) found a higher percentage (89.5%) of *Salmonella* sp. resistance to nalidixic acid. Resistance to nalidixic acid is the marker for decreasing susceptibility to fluoroquinolones which has been previously described for different serotypes (Whichard et al. 2007). Although fluroquinolones, specially ciprofloxacin resistance was not observed in the present study but resistance to this antibiotic in *Salmonella* is reported elsewhere (Lettini et al. 2016). Resistances to trimethoprim-sulfamethoxazole among NTS isolates from poultry and poultry meats were reported from Ghana (sulfamethoxazole—42·1%; trimethoprim—29·8%) (Andoh et al. 2016), Uganda (57.4%) (Odoch et al. 2018) and Egypt (83.3%) (Gharieb et al. 2015). It is a common antimicrobial used in poultry farms in Bangladesh (Imam et al. 2020). Considerable percentage of resistance to this antimicrobial in our study indicates we need to be careful using this one from now on. Otherwise, we will lose a popular drug of choice for chicken in near future.

Almost 79% of broiler liver and 62% of broiler meat had tetracycline residue in Bangladesh (Neogi et al. 2020). This represents the regular use of tetracycline compounds in broiler farms. Tetracycline resistance is mediated by *tet*A, *tet*B, *tet*C, *tet*D *and tet*G genes in Gram-negative bacteria. Moreover, bla_{TEM} and *tet*A genes are the most commonly found resistance gene in *Salmonella enterica* isolates (Pande et al. 2015), which we identified in the 23 isolates. Only six of the ampicillin-resistant isolates contained bla_{TEM} gene. Most of the tetracycline resistant isolates harbored the *tet*A gene, and none of the isolates contained the *tet*B and *tet*C genes. A study from India found the bla_{TEM} gene in five of 31 isolates and the *tet*A gene in nearly all isolates. They did not detect the *tet*B gene also (Waghamare et al. 2018). However, primers only for bla_{TEM}, *tet*A, *tet*B and *tet*C gene were available at the time of this study so resistant isolates could only be screened for these genes. Sequencing of the resistant genes was also not possible due to financial constraints. Future studies should be conducted to screen for other *S. enterica* genotypes and whole genome sequencing should be done for better understanding of resistance phenotypes of the bacteria.

A limitation of this study is the variability of sample sizes among hatcheries and that this study was only conducted in one region of Bangladesh. Thus, the results of this study might have limited external validity regarding the status of *Salmonella* across all poultry hatcheries in Bangladesh. Although serotyping is a widely used method all over the world for primary characterization of *Salmonella*, due to fund, and resource constraints we were not able to perform this, and other typing methods such as pulsed-field gel electrophoresis (PFGE), Random Amplified Polymorphic DNA (RAPD) for the isolates in the present investigation.

Conclusion

The present study reports the presence of motile *Salmonella* for the first time in chicken broiler breeder hatcheries in Chattogram, Bangladesh. As *Salmonella* is a food borne pathogen, the identified motile *Salmonella* has public health importance. Moreover, we observed antimicrobial resistance among the isolates. Therefore, proper steps should be taken to improve biosecurity and management of those hatcheries to reduce *S. enterica* infection as well as their resistance pattern. We should also think about antimicrobial stewardship program in Bangladesh perspective to minimize the antimicrobial resistance. Besides, the source of *Salmonella* is important to know for which further detailed epidemiological studies are needed. Further investigation is required to unveil the serotype of the isolates which originated from poultry hatcheries as well as to reduce the dissemination of resistant bacteria.

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Authors' contributions Conceptualization: Himel Barua; Methodology, Formal Analysis and investigation: Shafayat Zamil; Writing – original draft preparation: Shafayat Zamil; Writing – review and editing: Jinnat Ferdous, Mosammat Moonkiratul Zannat, Justine S. Gibson, Joerg Henning, Md. Ahasanul Hoque, Himel Barua; Funding acquisition: Himel Barua; Resources: Himel Barua; Supervision: Paritosh Kumar Biswas, Himel Barua.

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Availability of data and material Not applicable.

Code availability Not applicable.

Declarations

Competing interests The authors declare no competing interests.

Ethics approval As we collected only fluff samples from chicken hatcheries, we took verbal consent from the authority of the hatcheries.

Consent to participate Verbal consent from the authority of each hatchery was taken before collecting the samples from the hatcheries.

Consent for publication We give our consent for the publication of the submitted manuscript.

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